

# A380X OANS default airport

The moving airport map shows the default airport of the A380 FCOM without a manual selection

|          |                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------|
| Branch   | pr/25-a380x-oans-default-airport                                                                 |
| Aircraft | A380X (ND, OANS)                                                                                 |
| Commits  | 3a5afec9f feat(a380x/oans): display the FCOM default airport automatically                       |
| Base     | upstream master 2baa2b35e (independent)                                                          |
| Status   | CI checks pass; merged into develop a4e693727; tested in the sim on 2026-09-27 (user-confirmed). |

## Purpose

The OANS loaded the nearest airport on ground, but in flight only the destination (within 50 NM, at any height): no map after an air return or a diversion to the alternate. A manual selection stopped the automatic load for 10 minutes in every mode.

## Specification (A380 FCOM DSC-34-10-70-20, MOVING AIRPORT MAP and ARPT SEL)

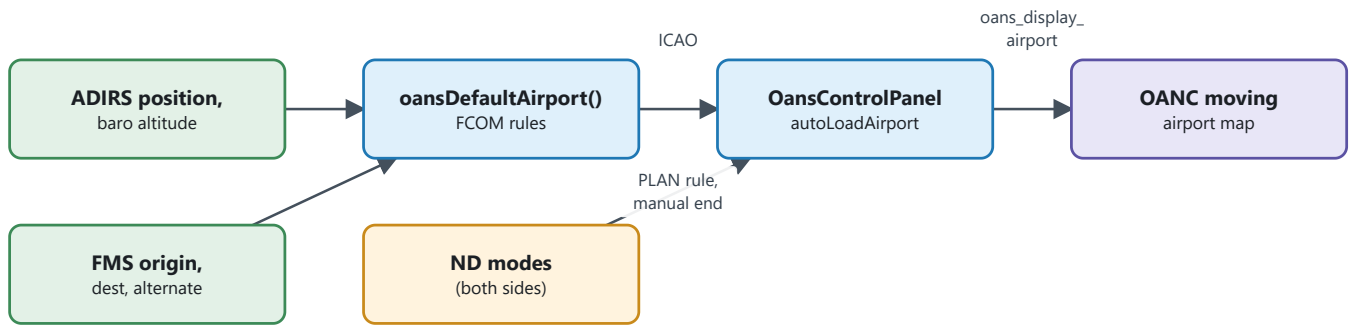
| ND mode           | Default airport                                                                                                                                                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ARC, ROSE-NAV     | on ground: the current airport; in flight: the origin, destination or alternate airport while the aircraft is within its virtual cylinder of 20 NM radius and 5000 ft height (the OANS keeps it)                                          |
| PLAN              | in flight: the origin airport closer than 50 NM to the aircraft, else the destination; the destination when the origin to destination distance is shorter than 300 NM; on ground the current airport (the map is centred on the aircraft) |
| Manual (ARPT SEL) | only the PLAN mode displays another airport (SET PLAN MODE in ARC / ROSE-NAV)                                                                                                                                                             |

## Design

- Pure function `oansDefaultAirport()` in `ND/OansDefaultAirport.ts` with the FCOM constants; `OansControlPanel.autoLoadAirport()` calls it every 2 s with the ADIRS position and baro corrected altitude (airport elevation in metres converted to feet) and the FMS origin, destination and alternate.
- In flight = FWC flight phases LIFT OFF to BELOW 800 FT (as before).
- One airport is displayed on both NDs (the synced `oans_display_airport` event): the PLAN rule applies when the other ND does not show ARC or ROSE-NAV, so the two sides never load different airports in turn.
- A displayed ARPT SEL airport is kept until the ND leaves the PLAN mode, or the aircraft lands or takes off (replaces the 10 minute lock and its FIXME).
- Unchanged: no automatic load without a present position, in the OANS performance mode out of the zoom range, while an airport selection is pending, or with the ARPT NAV reset panel pulled.

 Component added or changed by the feature    MSFS / simulator    Aircraft system (systems host, FMS, FWS)    Display / page

 Data, LVar, storage    Crew, flypad, external program



Architecture

## Tests

4 unit tests in `oansDefaultAirport.spec.ts`: on ground; the in-flight cylinder (radius, height, no valid altitude, a diversion to the alternate); keeping the displayed airport; the PLAN rules.

## Limitations

- The destination arrow (5 NM range, map out of range) is not part of this change.
- Both NDs share one displayed airport, where the real OANS has one per side.

## PR description (from the PR draft)

From `pr-drafts/25-a380x-oans-default-airport.md`, section "Summary of Changes" onwards.

### Summary of Changes

The OANS moving airport map displays the default airport of the A380 FCOM (DSC-34-10-70-20, MOVING AIRPORT MAP) without a manual selection. Before, the automatic load took the nearest airport on ground (kept), but in flight only the destination, within 50 NM at any height, and a manual selection stopped it for 10 minutes in every mode (the FIXME in `autoLoadAirport`).

- **ARC and ROSE-NAV:** the current airport on ground; in flight the origin, destination or alternate airport while the aircraft is within its virtual cylinder of 20 NM radius and 5000 ft height; the displayed one is kept while the aircraft stays in its cylinder.
- **PLAN:** in flight the origin airport if it is closer than 50 NM to the aircraft, else the destination airport; the destination when the origin to destination distance is shorter than 300 NM. One airport is displayed on both NDs (synced `oans_display_airport`), so this rule applies when the other ND does not show ARC or ROSE-NAV.
- **Manual selection (ARPT SEL):** only the PLAN mode displays another airport (FCOM: SET PLAN MODE in ARC / ROSE-NAV); a displayed airport is kept until the ND leaves the PLAN mode, or the aircraft lands or takes off.
- The rules are a pure function `oansDefaultAirport()` (`ND/OansDefaultAirport.ts`), used by `OansControlPanel.autoLoadAirport()` every 2 s with the ADIRS position and baro corrected altitude and the FMS origin, destination and alternate.
- 4 unit tests (on ground, the in-flight cylinder with a diversion to the alternate, keeping the displayed airport, PLAN).

### Cockpit API Changes

None.

### Screenshots (if necessary)

**TO ADD:** the OANS in ARC mode on approach to the destination, and after a diversion to the alternate.

### References

- A380 FCOM DSC-34-10-70-20 (Airport navigation, controls and indicators): MOVING AIRPORT MAP (default airport in ARC / ROSE-NAV and PLAN mode), ARPT SEL panel (SET PLAN MODE).

Discord username (if different from GitHub): **TO ADD**

### Testing instructions

1. On ground: the OANS displays the airport of the aircraft.
2. In flight with origin, destination and alternate in the F-PLN: the destination appears within 20 NM and 5000 ft of it; divert to the alternate: its map loads within 20 NM and 5000 ft.
3. ARPT SEL: display another airport in PLAN; back to ARC: the default airport returns.
4. PLAN in flight (other ND in PLAN too): the origin within 50 NM, then the destination.
5. `npx vitest run fbw-a380x/src/systems/instruments/src/ND/OansDefaultAirport.spec.ts` (4 tests).